



Warsaw University of Technology Team

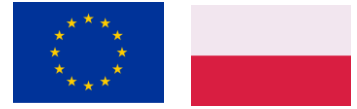
IFEC 2025 Social Event
Taoyuan, 16 July 2025



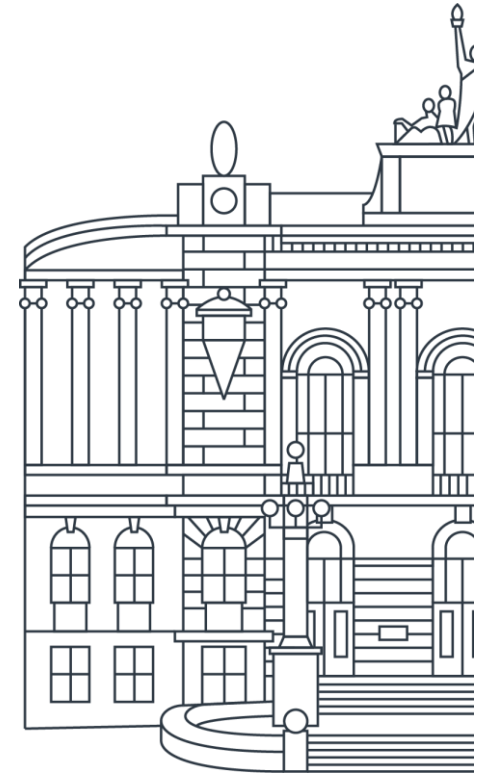
Faculty of Electrical Engineering

Warsaw University of Technology

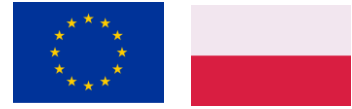
Agenda



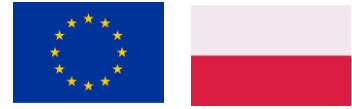
1. University introduction
2. Lab introduction
3. Challenges & solutions
4. Most memorable experiences
5. Questions for Mentors



Location: Poland, Warsaw



Warsaw University of Technology



📍 Established: 1826

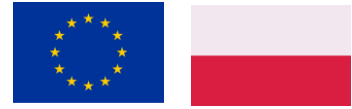
🎓 Faculties: 19

🌍 Students: 30,000

- Top-ranked technical university in Poland
- Over 600 international partnerships



Research team introduction



Power Electronics Laboratory of Warsaw University of Technology

Head of the laboratory (**Prof. Jacek Rąbkowski**) and the team



Krzysztof
Kalinowski MSc



Jan
Sitnik MSc



Prof. Jacek Rąbkowski



Rafał
Miśkiewicz PhD



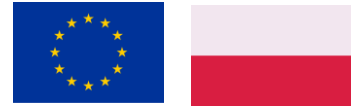
Rafał
Kopacz PhD



Przemysław
Trochimiuk PhD



Research focus



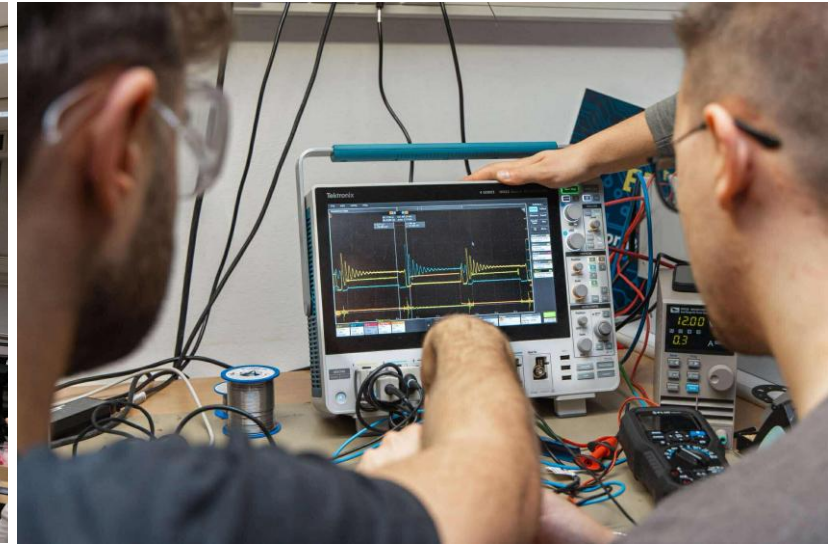
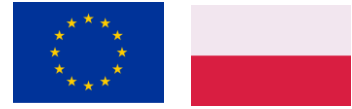
- Next-Generation Power Converters:
 - Design of modular, reconfigurable, and bidirectional (V2G) EV charging systems
 - High-power isolated chargers with advanced control,
 - Energy management in microgrids and prosumer applications
- Electric Vehicle Charging Systems
- Integration with Renewable Energy and Energy Storage



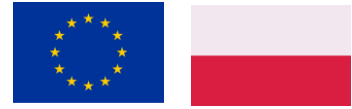
MoReSiC - *Modularized, Reconfigurable and Bidirectional Charging Infrastructure for Electric Vehicles with Silicon Carbide Power Electronics*



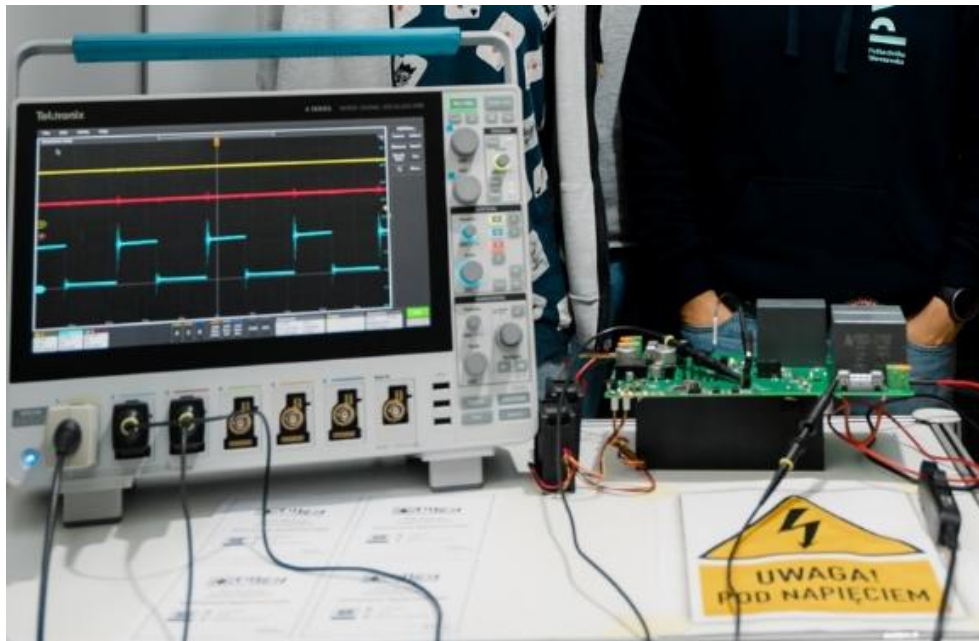
Lab introduction



Students' research groups introduction

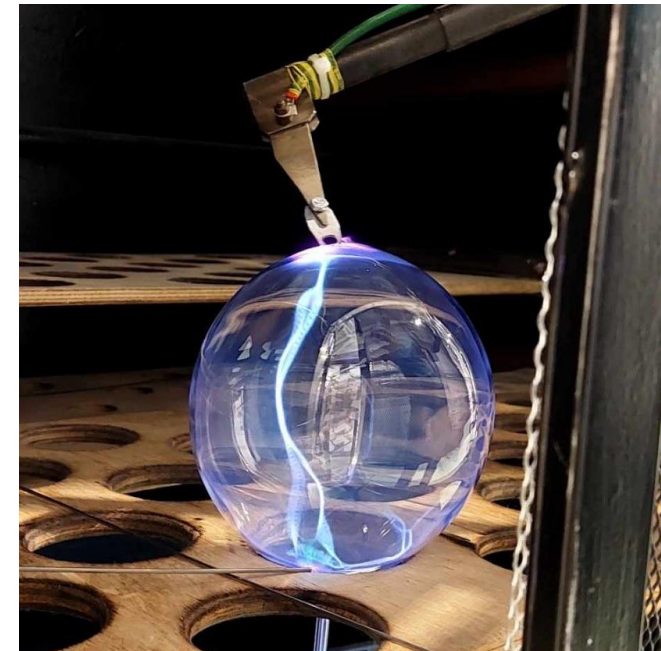


Drones Renewables Research Group



Drone charging station

Power Electronics And Plasma Applications Research Group



Plasma ignition with a power converter



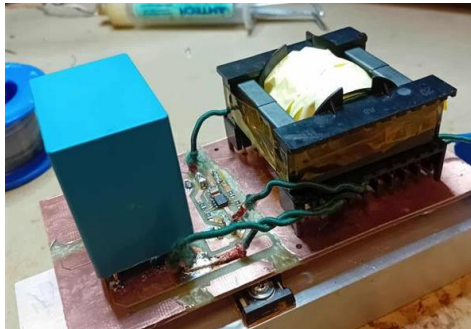
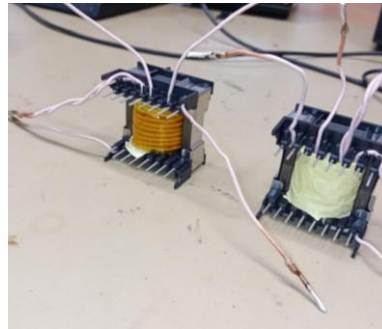
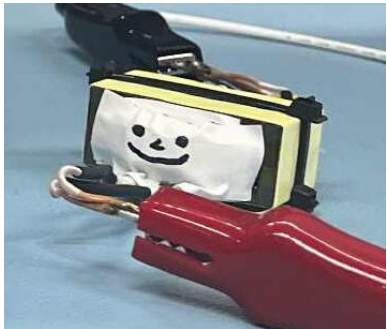
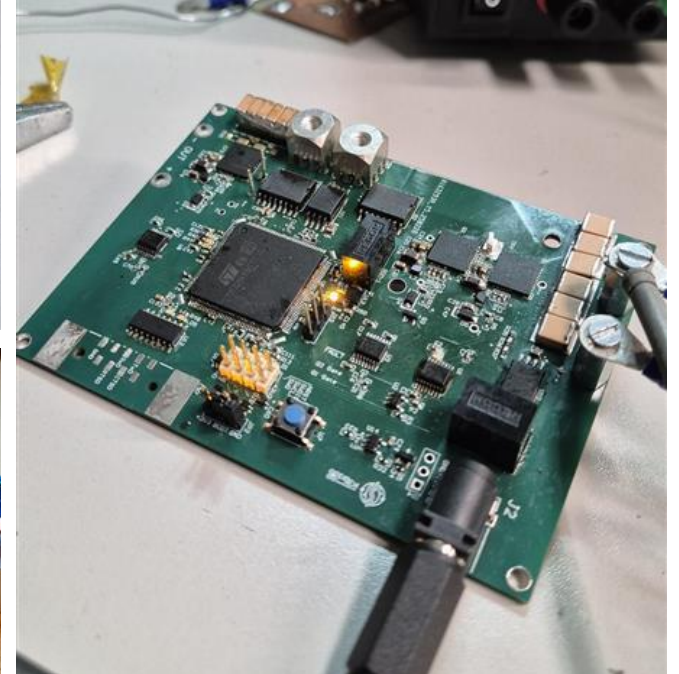
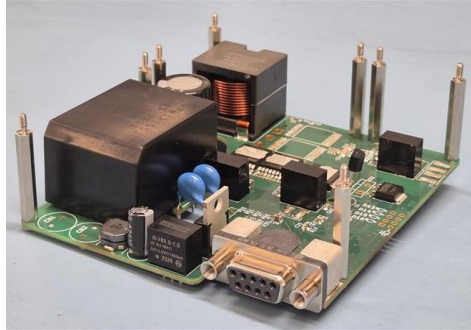
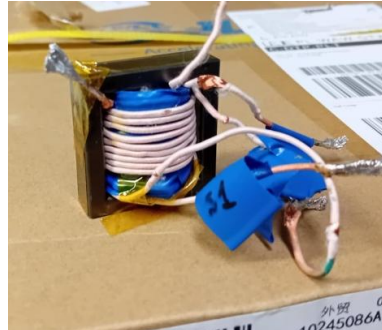
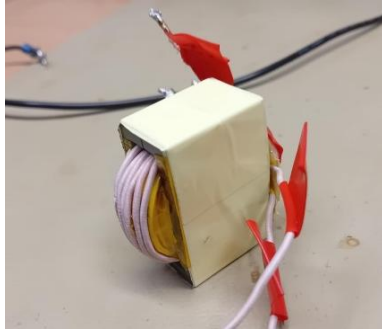
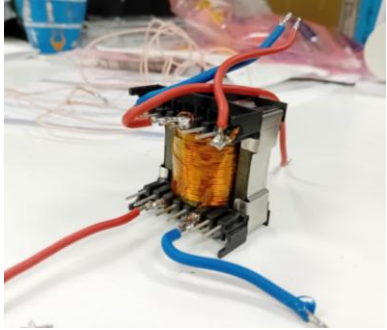
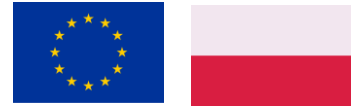
IFEC team origins



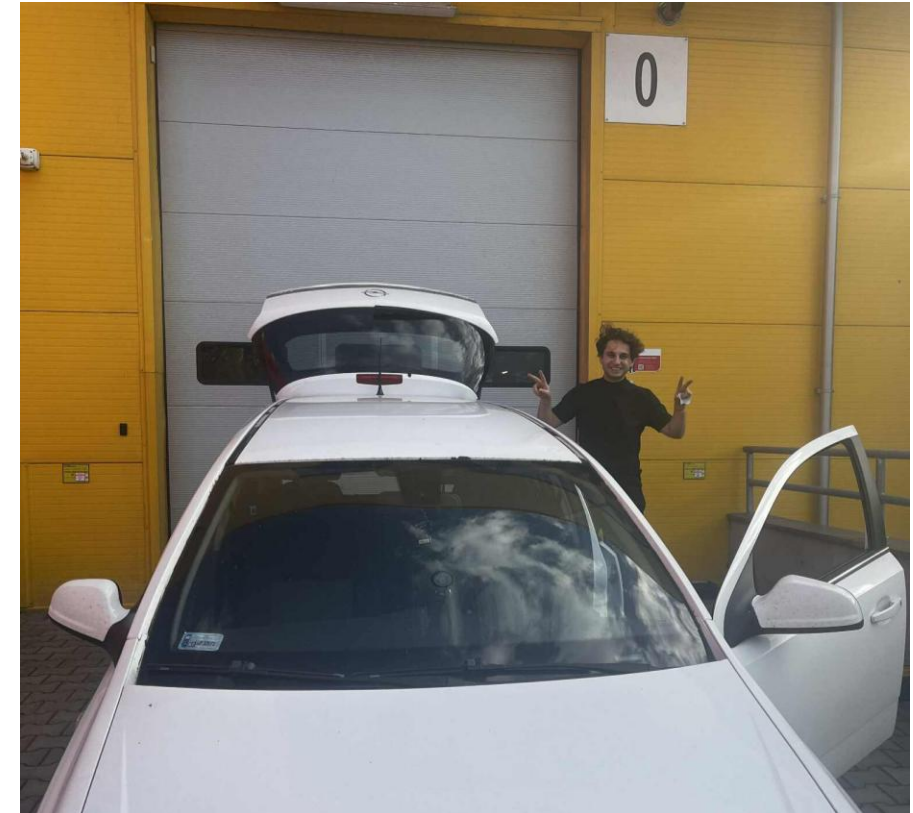
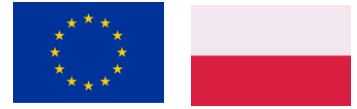
Visit in the Institute of Plasma Physics and Laser Microfusion



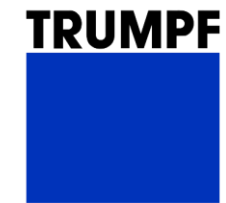
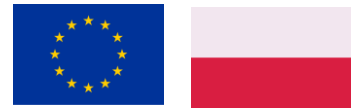
Challenges & solutions – many prototypes



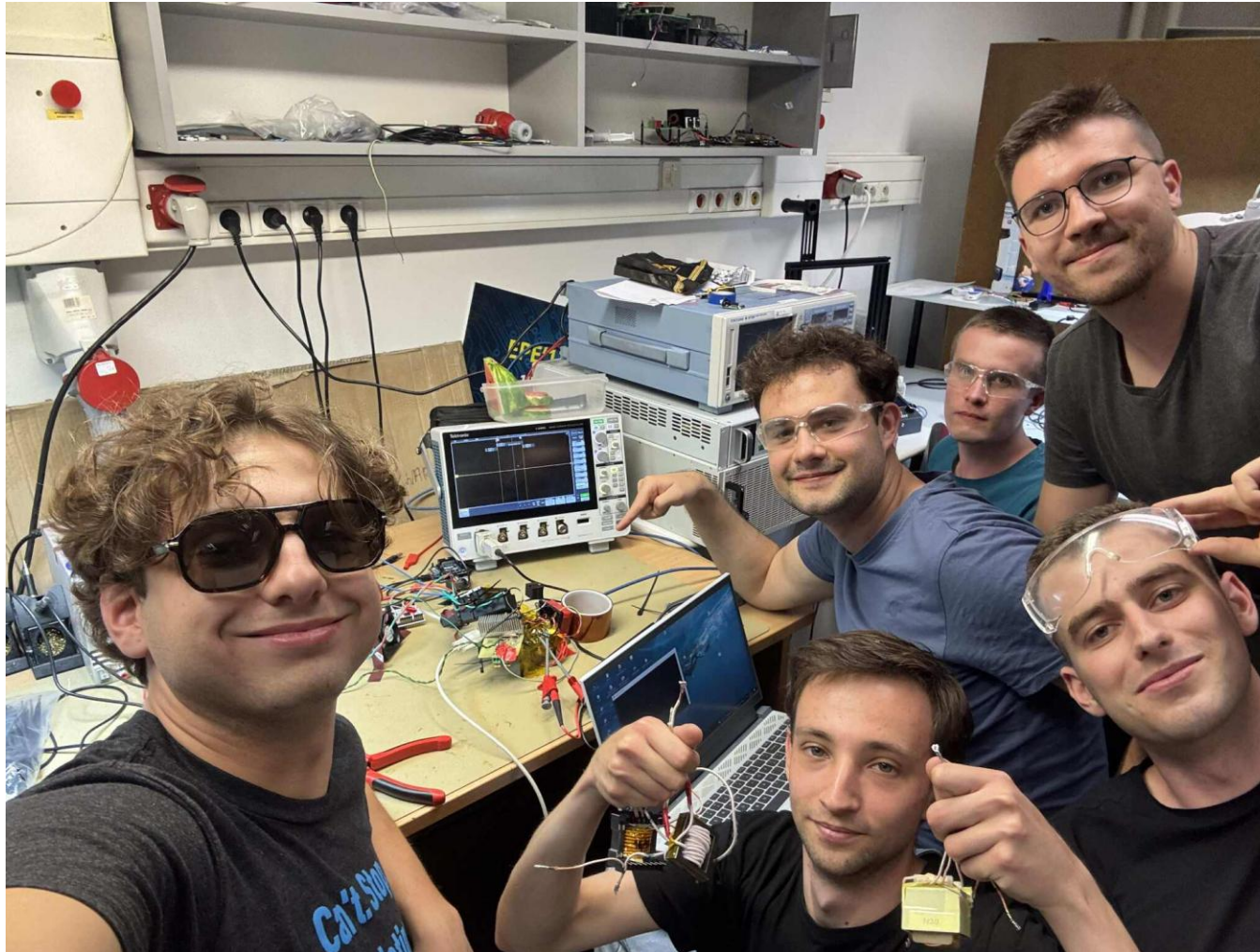
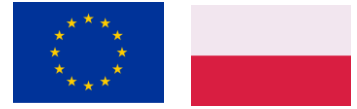
Challenges & Solutions – lost PCBs



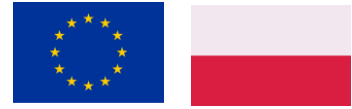
Challenges & Solutions - funding



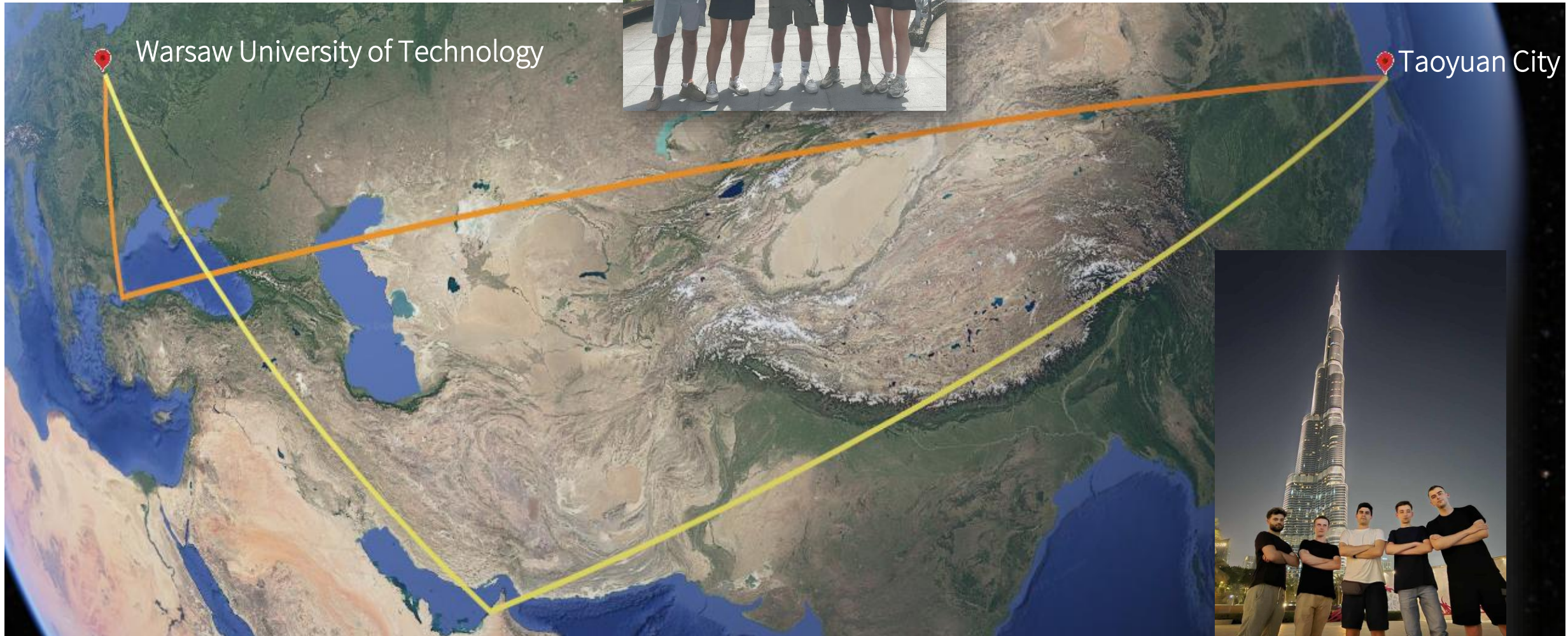
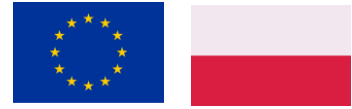
Most memorable experiences – first working prototype



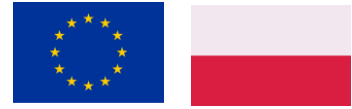
Most memorable experiences - qualifying for the final



Most memorable experiences



We made it!





Thank you for your attention

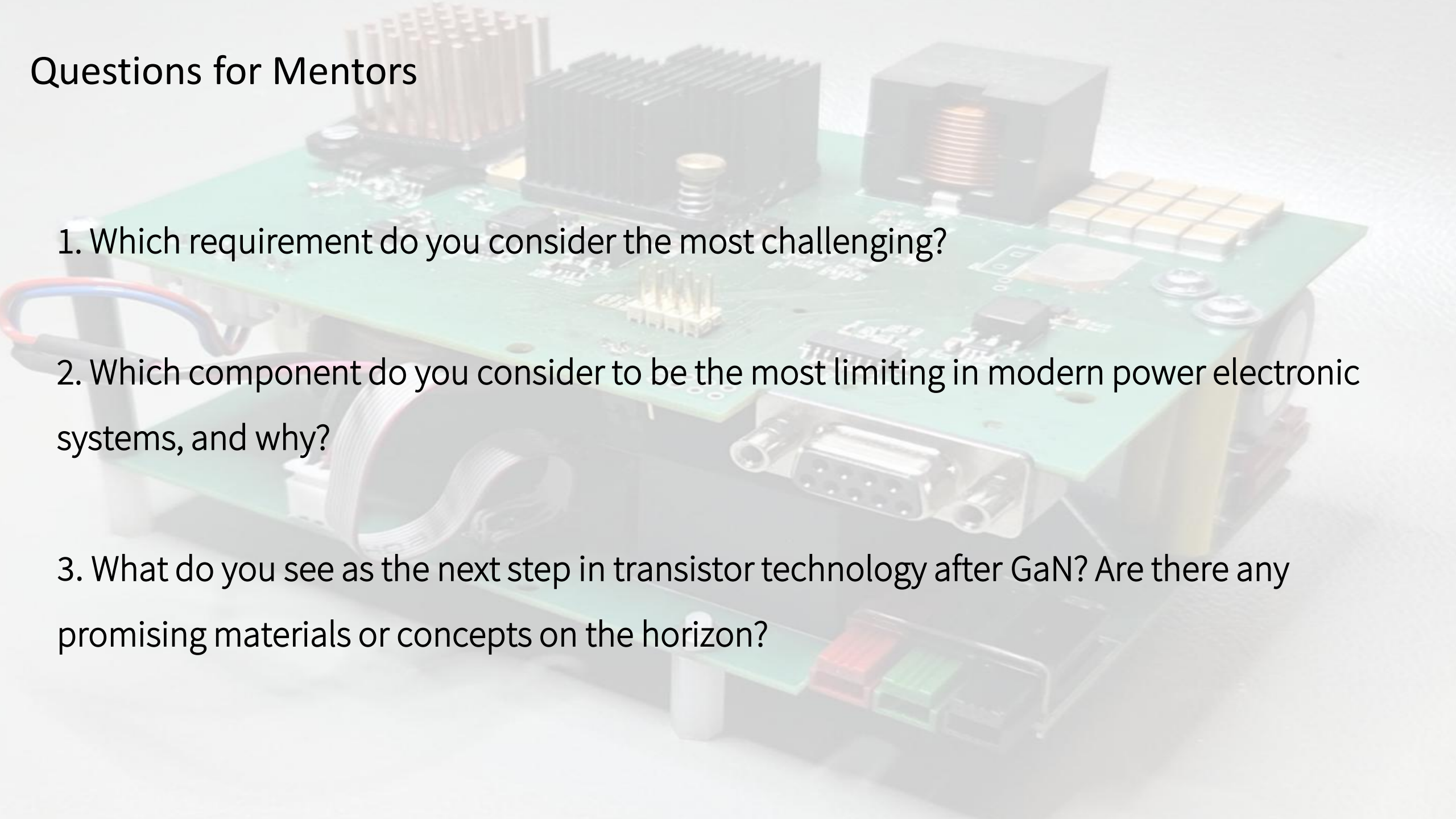
Polish Team
IFEC 2025



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Warsaw University of Technology

Questions for Mentors



1. Which requirement do you consider the most challenging?
2. Which component do you consider to be the most limiting in modern power electronic systems, and why?
3. What do you see as the next step in transistor technology after GaN? Are there any promising materials or concepts on the horizon?

Our partners



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